

Engineering Notes

Forces and Moments Generated by Swept-Forward Grid Fins and Planar Fins

Marco Debiasi*

*Centre for Defence Engineering, Cranfield University,
Shrivenham, England SN6 8LA, United Kingdom*

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Nomenclature

C_F	=	force coefficient
C_M	=	moment coefficient
c	=	grid-fin chord (lattice element chord), mm
D	=	diameter of ogive-cylinder body, mm
h	=	grid-fin height, mm
L	=	length of ogive-cylinder body, mm
M_∞	=	freestream Mach number
p	=	static pressure, Pa
R	=	radius of curvature of ogive nose, mm
Re_D	=	Reynolds number based on body diameter
s	=	fin span, mm
T	=	static temperature, K
t	=	planar fin thickness, mm
U_∞	=	freestream velocity, m/s
w	=	thickness of walls of lattice cells, mm
x	=	longitudinal coordinate of vehicle
y	=	spanwise coordinate of vehicle
z	=	normal coordinate of vehicle
α	=	angle of attack, deg
Λ	=	sweep angle of the grid fin lattice, deg

I. Introduction

GRID fins are unconventional control surfaces originally developed in Russia [1], the geometrical configurations and aerodynamic properties of which have been described in theoretical [2–6], experimental [7–24], and numerical [25–45] studies. They have a smaller chord than planar fins, so they generate smaller hinge moments [7], which require smaller actuators to rotate them in a high-speed flow. Their small chord also makes them less likely to stall at high angles of attack, which increases their control effectiveness [7]. Finally, they can be conformally folded over an aerodynamic body to create a compact and convenient package to store.

Grid fins can have higher or lower drag than planar fins, depending on the speed of the airflow [13,14]. They have a disadvantage in the transonic regime in which the flow chokes in the cells of their lattice, thus reducing the flow rate through the fin, which effectively acts as

an obstacle to the flow [34,35]. At low subsonic and at supersonic speeds, the lattice manages to swallow the flow, thus reducing the drag and improving the control characteristics.

This study complements and expands one previously conducted on the forces and moments generated by globally swept-back grid fins [24]. A globally swept-forward grid fin with sharp leading edges rather than a swept-back geometry has been considered and should offer similar aerodynamic characteristics. Swept-forward grid fins would be easier to deploy from their folded position over a vehicle's body since, unlike swept-back fins that need to rise up from the stored to deployed position against the aerodynamic forces, swept-forward fins would deploy with the help of the aerodynamic forces. Additionally, this study directly compares the aerodynamic characteristics of swept-forward grid fins to those of planar fins. To this aim, aerodynamic data have been obtained for a representative planar-fin geometry. Some data have been obtained anew for the reference grid-fin configuration [24] to verify their consistency and comparability. Wind-tunnel measurements were conducted at freestream Mach numbers ranging from 0.72 to 1.30 and at angles of attack up to 12 deg at selected Mach numbers.

II. Experimental Setup

The experimental setup is the same as that of Debiasi and Zeng [24] with stainless-steel models in a transonic wind tunnel.

The geometry of the fins is presented in Fig. 1, in which all the dimensions are relative to the diameter $D = 114.3$ mm of the ogive-cylinder body on which the fins were mounted. This is a $7D$ -long cylindrical body capped by a $3D$ -long tangent ogive nose. The grid fins have a rectangular outer frame with span s , height h , and chord c of size $0.75D$, $0.333D$, and $0.118D$, respectively. The thickness w of the walls of the lattice is $0.007D$. Figures 1c and 1d show the top view of the baseline (reference) and of the globally swept-forward grid fins, respectively. The baseline grid fin has the same geometry discussed in numerous studies [5,10,11,13,14,16,21,23–28,34–37,42]. The geometry of the swept-forward fin is obtained by tilting forward the framework of grid cells by an angle $\Lambda = 30$ deg while maintaining the same projected structure and dimensions of the baseline fin shown in Fig. 1b. The leading edges of the lattice walls of the swept-forward fin have a 20 deg sharp profile (SF sharp) to improve the flow ingestion at transonic and supersonic speeds [19,21,23,24,39]. The planar fin was not designed to match the aerodynamic characteristics of either type of grid fin at some specific flow conditions. Rather, a representative clipped-delta planar fin that has the same span s of the grid fins, a root chord equal to the span, a tip chord of half this value, and thickness t equal to $0.048D$ was designed, Fig. 1e. Four fins mounted $1.5D$ forward of the body base were used in all the measurements. Figure 2 shows a picture of the rear part of the body with concurrent installation (not tested) of the baseline grid fin, the planar fin, and the SF-sharp grid fin, clockwise from left to right.

The ogive-cylinder body was installed on a sting in the center of the wind-tunnel test section. A balance with resolution of 0.014 N and accuracy of 0.092% located close to the center of gravity of the body was used to measure the forces and moments acting on it. Its coordinate system had the x axis in the body longitudinal direction, the y axis in the spanwise direction, and the z axis normal to these. The data were recorded simultaneously to the total and static values of the pressure and temperature in the test section and corrected for the effect of the pressure measured at the base of the body.

Measurements were taken at freestream Mach numbers M_∞ between 0.72 and 1.30. The test section of the blowdown wind tunnel has slanted circular perforations with suction capability for reducing wall interference and shock reflections. A flow control

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*Senior Research Scientist, Temasek Laboratories, National University of Singapore; currently Research Fellow; marco.debiasi@cranfield.ac.uk. Member AIAA.

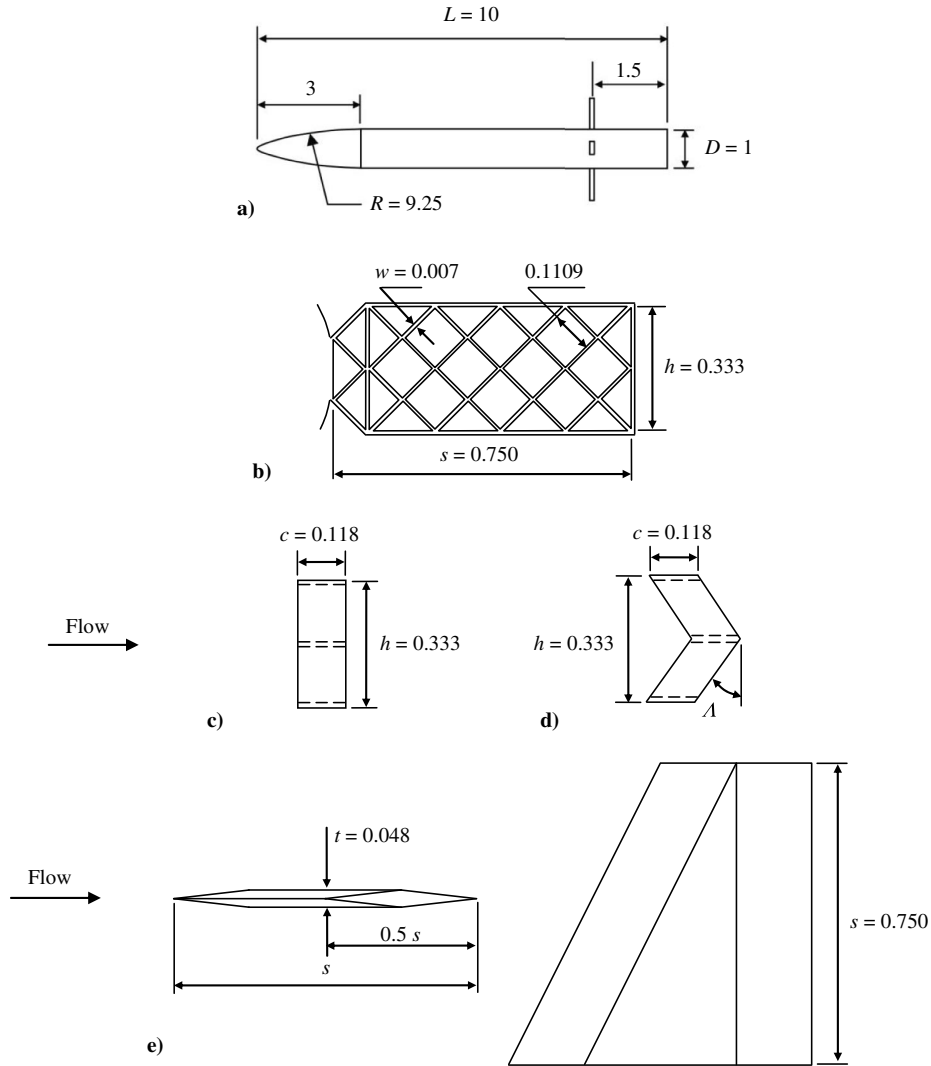


Fig. 1 Schematic of the models for wind-tunnel testing: a) ogive cylinder with grid fins, b) front view of the grid-fin structure, c) top view of the baseline, d) top view of the swept-forward grid fin, and e) top and side views of the planar fin.

system maintains the Mach number and the pressure in the test section within 1% of the desired values during the data acquisition. The corresponding static pressure p and the static temperature T are indicated in Table 1.



Fig. 2 Picture of the rear part of the ogive-cylinder body with concurrent installation of the baseline grid fin, the planar fin, and the swept-forward grid fin (clockwise from left to right). In the wind tunnel, the holes of the screws were covered by plugs to maintain an aerodynamically smooth surface.

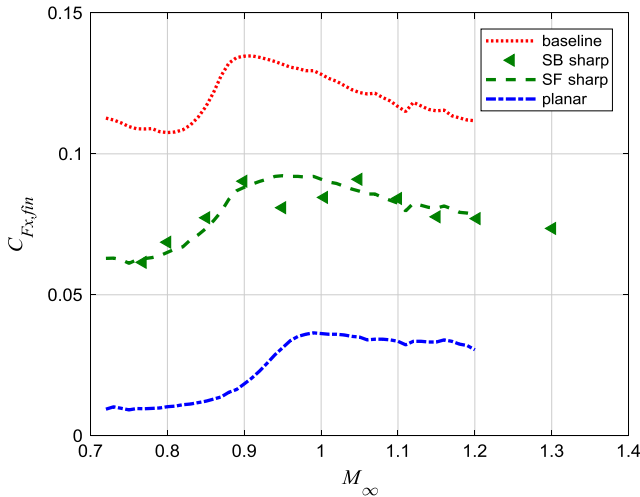
III. Results

The force and moment coefficients were obtained from the measured values by using the body diameter and the area of its base as dimensional references. No significant differences were found between repeated measurements. The incremental forces produced by the fins were obtained as the difference of those of the body with and without fins at the same flow conditions.

Axial-force (drag) measurements were taken by varying M_∞ between 0.72 and 1.20 (high-subsonic and transonic flow conditions) with the body at an angle of attack α equal to zero. Figure 3 presents the incremental axial-force coefficient $C_{F_{x,fin}}$ of a single fin. The typical drawback of grid fins, namely, their high drag at transonic speed, is well evidenced in the figure. The baseline grid fin has substantially higher drag than the planar fin with Mach number variations resembling those observed for similar [11,13,14] or comparable [7,15] fins. The SF-sharp fin reduces the drag by 30 to 35% relative to the baseline, a result analogous to that of a globally swept-back grid fin with sharp leading edges (SB sharp) [24]. Wang and Yu [41] obtained larger drag reductions with numerical simulation of individual grid cells of the same geometry. The axial-force coefficient of the planar fin increases with the freestream velocity and peaks at $M_\infty = 1.00$, as expected for a wing in transonic flow. The coefficients of the baseline and of the SF-sharp grid fins also increase with the freestream velocity but peak at $M_\infty = 0.90$ and $M_\infty = 0.95$, respectively. This is associated with the flow choking in the lattice, which is more severe for the baseline fin

Table 1 Flow conditions in the wind-tunnel test section

M_∞	p , Pa	T , K	U_∞ , m/s	$Re_D \times 10^6$
0.70	$1.14 \cdot 10^5$	274	232	2.24
0.80	$1.04 \cdot 10^5$	266	261	2.49
0.90	$9.38 \cdot 10^4$	256	289	2.74
1.00	$8.34 \cdot 10^4$	247	316	2.89
1.10	$7.45 \cdot 10^4$	238	340	2.97
1.20	$6.55 \cdot 10^4$	228	362	2.99
1.30	$5.73 \cdot 10^4$	224	389	2.88

**Fig. 3** Fin incremental axial-force coefficient vs freestream Mach number at $\alpha = 0$ deg (SB-sharp data from Ref. [24]).

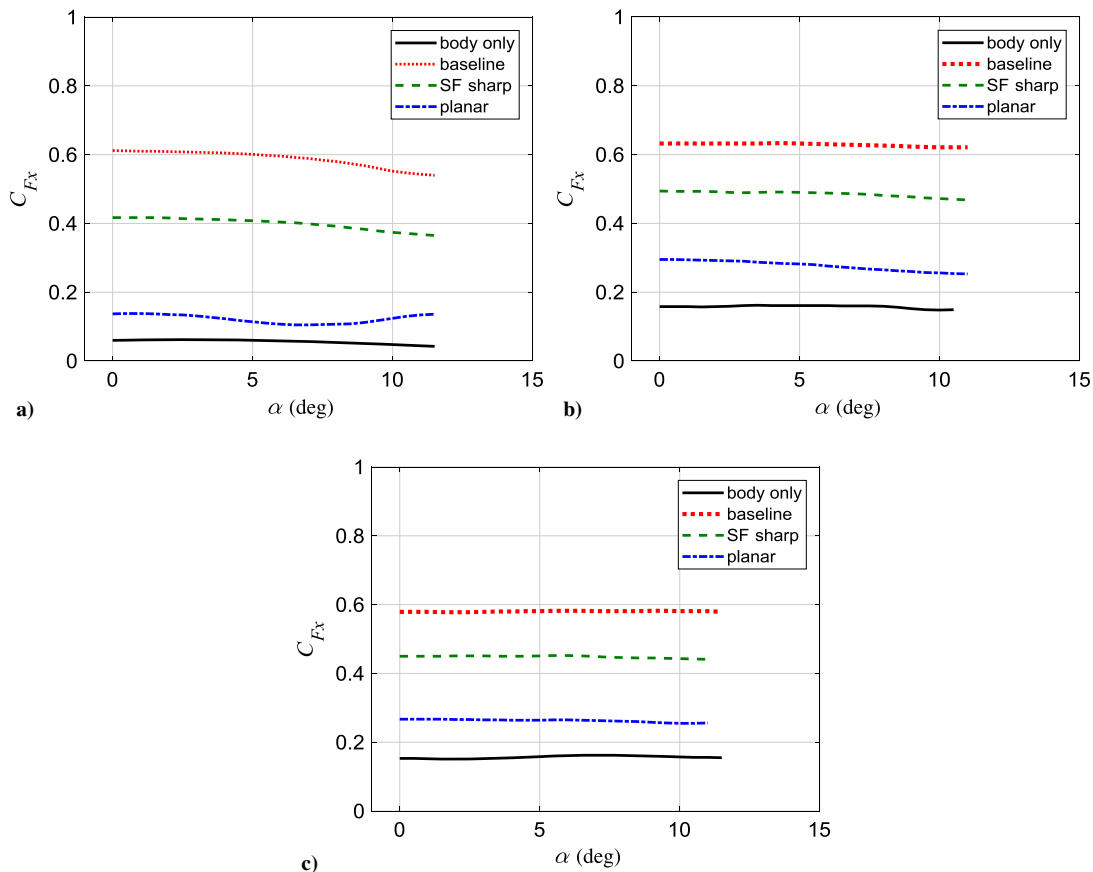
than for the SF-sharp fin. The axial-force coefficient of all the fins starts decreasing above $M_\infty = 1.10$.

The effect of the angle of attack α of the body was explored by varying it from 0 to 12 deg at $M_\infty = 0.90$ and 1.10, which are considered to be significant transonic conditions, along with the moderate supersonic value of $M_\infty = 1.30$. Measurements, not shown here, were performed by varying α from 0 to -12 deg to verify that the results mirrored those obtained at positive angles.

Figure 4 shows the axial-force coefficient of the body alone and with fins. In all cases, the coefficient varies little with the angle of attack. The body with baseline grid fins has a coefficient of about 0.6, close to the results reported for this or comparable fins in other studies [13,14,29,30]. The drag reduction of the four SF-sharp grid fins relative to the baseline fins is consistent with that of two lateral (elevator) SB-sharp fins shown in Ref. [24]. Planar fins have lower drag than the SF-sharp fins in these flow conditions.

Figure 5 shows that the normal-force coefficient of the body without and with fins increases with the angle of attack. The results are similar to those available in the literature for the body without and with comparable fins [8,30,33]. The contribution of the vertical (rudder) planar fins is deemed negligible, whereas the vertical grid fins moderately add to the normal force produced by the lateral grid fins [13,25,28,30]. At $M_\infty = 0.90$, the normal force of the body with baseline fins drops somewhat for $\alpha > 8$ deg, an effect also observed at $\alpha \approx 7$ deg using two lateral baseline fins [24]. Similar to their swept-back counterparts, the SF-sharp fins do not create such a drop. At $M_\infty = 1.10$ and 1.30, the normal-force coefficient with the baseline and the SF-sharp fins are practically equivalent, the latter fins having a marginal benefit. At these Mach numbers and angles of attack, the planar fins have larger normal-force coefficient.

This is also reflected in the pitching-moment coefficient of the body, Fig. 6, which is linearly related to the corresponding normal-force coefficient. Since the fins are located at the tail of the body, their positive normal force produces a negative pitching moment relative to the balance, which is located close to the center of gravity of the

**Fig. 4** Body axial-force coefficient vs angle of attack at a) $M_\infty = 0.90$, b) $M_\infty = 1.10$, and c) $M_\infty = 1.30$.

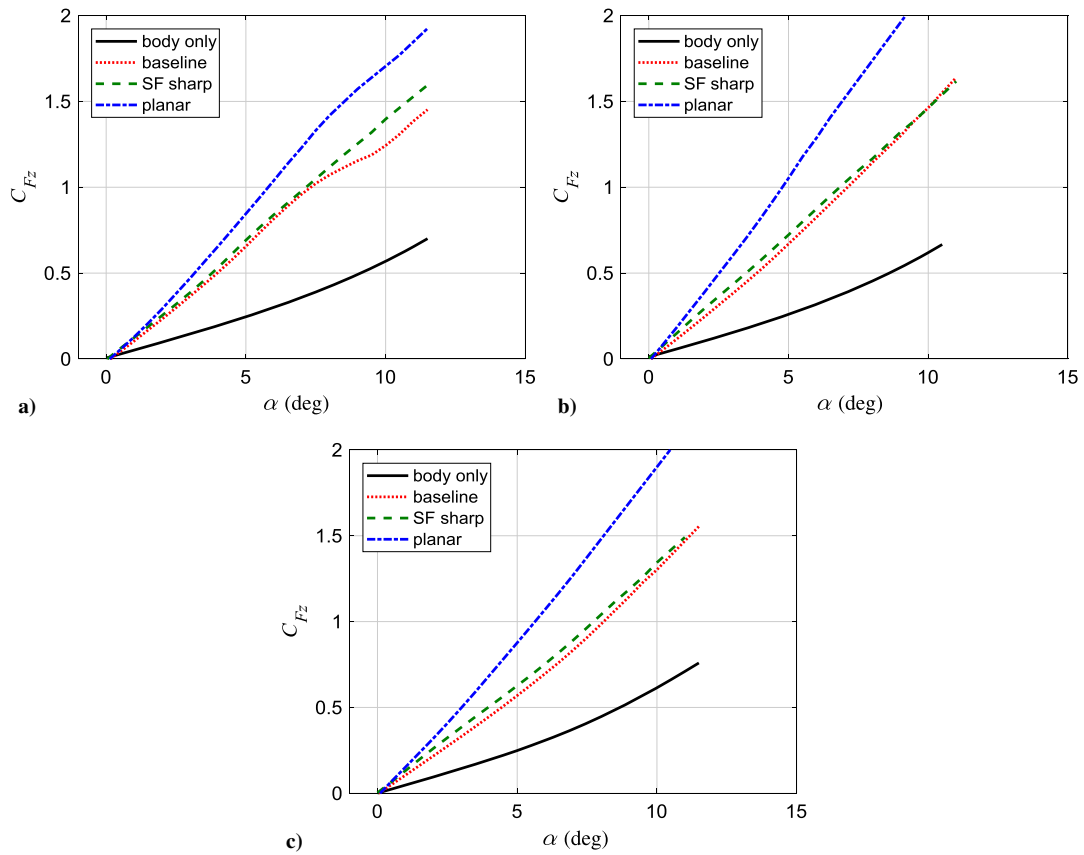


Fig. 5 Body normal-force coefficient vs angle of attack at a) $M_\infty = 0.90$, b) $M_\infty = 1.10$, and c) $M_\infty = 1.30$.

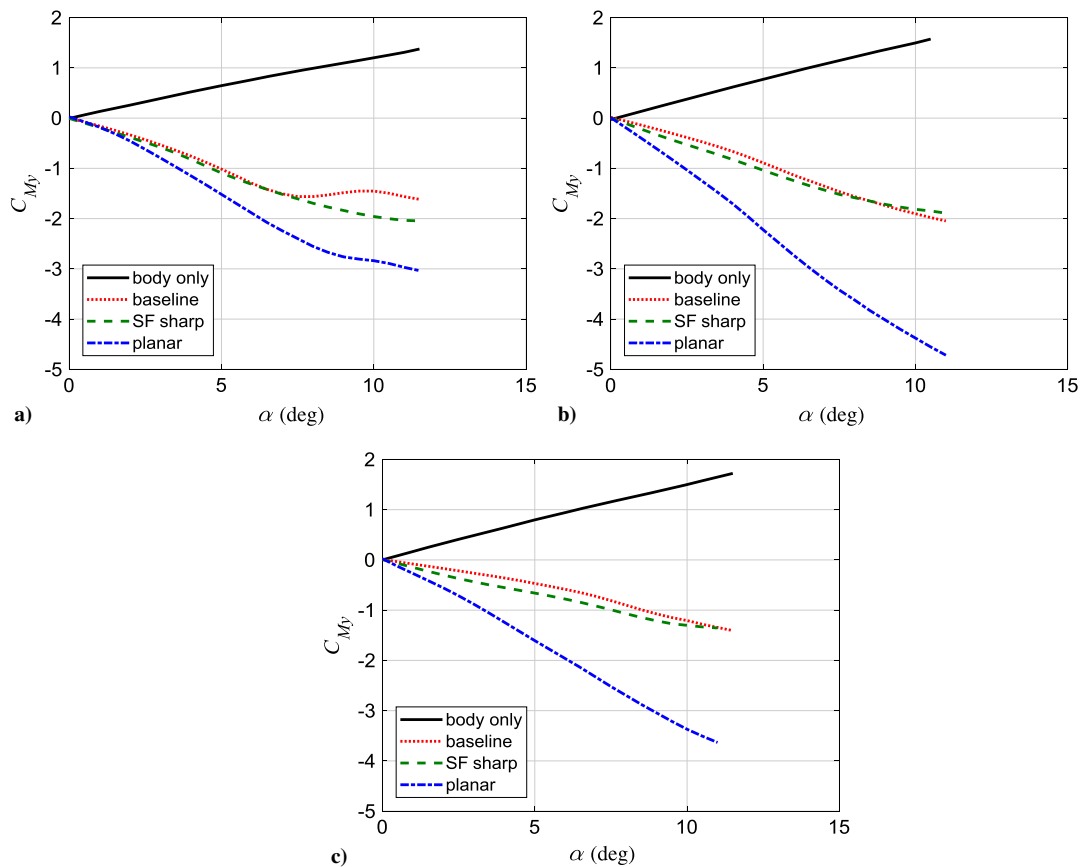


Fig. 6 Body pitching-moment coefficient vs angle of attack at a) $M_\infty = 0.90$, b) $M_\infty = 1.10$, and c) $M_\infty = 1.30$.

body. Unlike their swept-back counterparts, the SF-sharp fins create only marginally larger (but at $M_\infty = 0.90$ smoother) pitching moments than the baseline fins. In all cases, the values are smaller than those of the planar fins, indicating that the latter have better control characteristics for maneuvering a vehicle at transonic and moderately supersonic speed.

IV. Conclusions

Measurements have been conducted at transonic conditions and in a range of angles of attack of the aerodynamic characteristics of grid fins without and with a globally swept-forward lattice structure having sharp leading edges. These are compared to the measurements of representative planar fins in the same flow conditions. All types of fins were mounted on an ogive-cylinder body installed in a blowdown wind tunnel. The results obtained indicate that swept-forward grid fins with sharp leading edges have performance comparable to that of similarly featured swept-back grid fins. While both types of globally swept grid fins have better aerodynamic characteristics than a much-referenced unswept type, their performance falls short of that of planar fins in the range of speeds and angles of attack explored. These results, together with others available in the literature for various grid-fin geometries, suggest that globally sweeping the lattice provides moderate aerodynamic benefits in the transonic regime and that increasing a grid-fin performance entails optimizing the profiles of its lattice walls. Additional research is required to understand if concurrent optimization of the walls' profiles in a globally swept grid fin can produce aerodynamic characteristics competitive with those of planar fins.

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